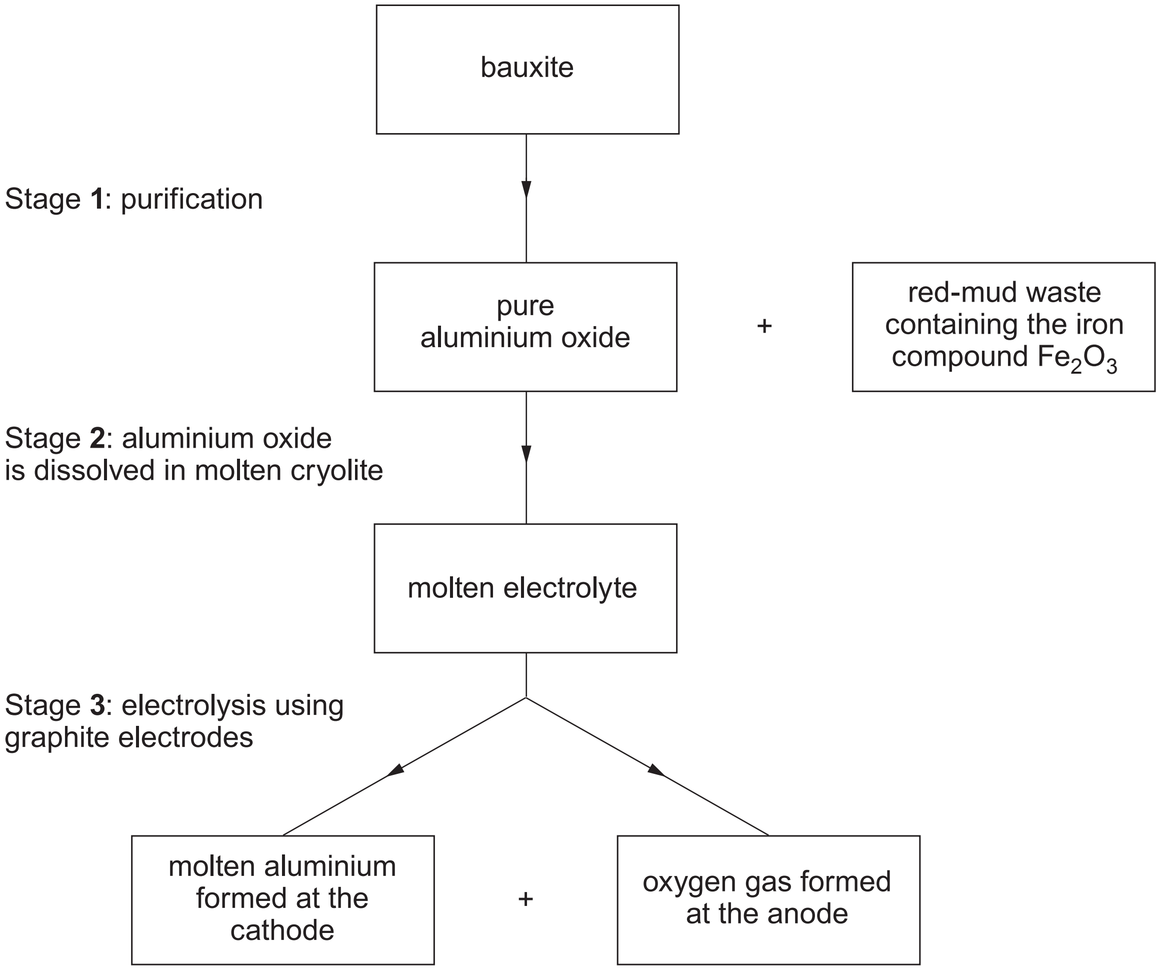
5. (a) The flow chart shows the stages involved in the extraction of aluminium from bauxite. Bauxite is a mixture of mainly aluminium oxide and a compound of iron.



Examiner only

(i) Underline the chemical name for the iron compound, Fe_2O_3, found in the red-mud waste. [1]

iron(V) oxide **iron(II) oxide** **iron(III) oxide** **iron(I) oxide**

(ii) Give **one** reason why aluminium oxide is dissolved in molten cryolite in stage **2**. [1]

....

....

(iii) Explain why the anode is regularly replaced in stage **3**. [2]

....

....

(iv) Complete and balance the equation for the electrolysis of aluminium oxide. [2]

$$2Al_2O_3 \longrightarrow \boxed{}\,Al + 3 \ldots\ldots\ldots$$

(b) The recycling rate for aluminium drinks cans in the UK reached 72 % in 2017.

Suggest why most aluminium waste is recycled. [2]

....

....

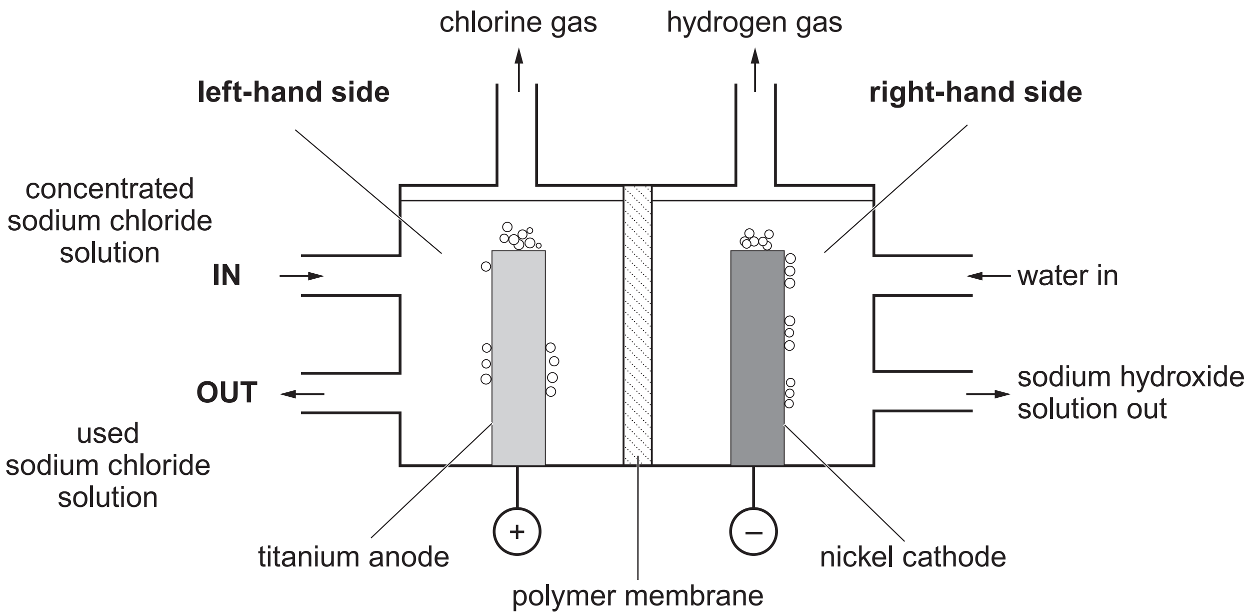
....

....

(c) The diagram shows a model of a membrane cell used to manufacture sodium hydroxide solution, chlorine and hydrogen by the electrolysis of aqueous sodium chloride.

The membrane cell is designed so that chlorine is kept separate from hydrogen and aqueous sodium hydroxide.

The polymer membrane only allows positive ions to pass through it. Sodium ions and hydrogen ions pass through from the left-hand side to the right-hand side of the cell.



Chlorine is an extremely corrosive substance which reacts with most metals. Titanium reacts quickly with chlorine gas but it resists attack by chlorine in aqueous conditions.

Examiner only

(i) Put a **tick** (✓) in the box next to the correct list of ions found on either side of the membrane during the process. [1]

Left-hand side of the membrane	Right-hand side of the membrane	
Na^+ Cl^- H^+ OH^-	Cl^- H^+ OH^-	☐
H^+ OH^- Cl^-	Na^+ H^+ OH^-	☐
Na^+ Cl^- H^+ OH^-	Na^+ H^+ OH^-	☐
Na^+ Cl^-	Na^+ OH^-	☐

(ii) Put a **tick** (✓) in the box next to the correct statement. [1]

- titanium burns in chlorine in aqueous conditions ☐
- titanium doesn't react with chlorine under any conditions ☐
- aqueous conditions prevent chlorine from reacting with titanium ☐
- aqueous conditions prevent chlorine from reacting with sodium chloride ☐

(iii) Put a **tick** (✓) in the box next to the two substances formed on the right-hand side of the membrane if the sodium chloride solution were contaminated with magnesium bromide. [1]

- sodium hydroxide and magnesium hydroxide ☐
- sodium hydroxide and sodium bromide ☐
- sodium hydroxide and magnesium bromide ☐
- magnesium chloride and sodium ☐

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